LEFA LAZY MALATJI

Software Developer | NLP Researcher

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Professional profile

Software developer and MSc Computer Science candidate with one year of internship experience developing web applications and managing PostgreSQL databases. Builds applications using C#, ASP.NET Core and PHP, supported by a Computer Science and Statistics background. Published NLP researcher investigating Sepedi-English text generation, with additional experience teaching programming and database concepts.

Professional experience

Lecturer | Rosebank College

Current position

- Teach programming, software development and database concepts through practical coding activities, including C#, object-oriented programming, HTML/CSS, ASP.NET MVC and SQL.
- Assess student projects against rubrics and provide technical feedback on implementation and problem-solving.

Software Developer Intern | Simplifon

One-year internship

- Developed and maintained web-based applications.
- Developed and managed PostgreSQL databases supporting web applications.

Academic Tutor | University of Limpopo

Three years

- Tutored Advanced Databases, Web Development and Programming Logic and Design; supported practical coding and problem-solving.

Selected projects

Project	Implementation and contribution
Store Hub	**PHP	PostgreSQL** Built a multi-role e-commerce platform with AI-driven personalisation, intelligent product search, business intelligence reporting and a session-aware support chatbot. Source code
HelpDeskHub	**C#	ASP.NET Core	PostgreSQL	Entity Framework** Developing a help desk application with ticket APIs, user identity and a relational data layer. Project in progress. Project overview

Project	Implementation and contribution
Portfolio Guide	**JavaScript | HTML | CSS | GitHub Pages** Built a browser-based guide with factual answers, project follow-up context, source links and contact options. Added automated checks and deployment through GitHub Actions. Live portfolio

Education

MSc in Computer Science | University of Limpopo

2026-present | In progress

Research focus: Investigation of data-scarcity effects for the development of a Sepedi-English text generation model, comparing FNet, LSTM and mBERT.

BSc Honours in Computer Science | University of Limpopo

2025

BSc in Mathematical Sciences | University of Limpopo

2021-2024 | Majors: Computer Science and Statistics

Technical skills

Software development: C#, ASP.NET Core Web API, MVC, PHP, JavaScript, HTML5, CSS3 and Bootstrap 5.

Database engineering: PostgreSQL, Microsoft SQL Server, MySQL, SQL, data modelling, relational schema design, normalisation and Entity Framework migrations.

AI and NLP: Python, TensorFlow/Keras, text preprocessing, language modelling, FNet, LSTM, mBERT and evaluation using BLEU and ROUGE.

Data analysis: Statistics, data preparation, descriptive analysis, SQL aggregation and interpretation of experimental results.

Tools and practices: Git, GitHub, Docker, REST APIs, Swagger and Agile development.

Research and publication

Lefa L. Malatji and Thipe I. Modipa. Investigating Data Scarcity and Model Adaptation for Sepedi-English Code-Switching Text-Generation.

- Co-authored published research on generative text modelling for Sepedi-English code-switching using the FNet architecture.
- Current MSc work examines data scarcity and compares language models for a low-resource South African language setting.

Languages

Sepedi and English